Cloud Computing & DevOps- Task 2

Containerization Using Docker and Deployment on a Cloud Virtual Machine

Objective

In **Task 1**, you learned how to:

- Host a static website using cloud storage (S3)
- Understand basic cloud hosting concepts
- Make your application publicly accessible

In **Task 2**, you will learn how **modern DevOps engineers package and deploy applications using containers.**

The objective of this task is to:

- Understand containerization concepts
- Create a Docker image for your website
- Run the application inside a container
- Deploy the container on a cloud-based virtual machine

This task introduces you to **real DevOps workflows used in production environments.**

Why This Task Matters (Industry Context)

In real-world systems:

- Applications are not deployed as loose files
- Different environments (local, staging, production) must behave the same
- Scaling and portability are critical

Docker solves these problems by:

- Packaging the application with its dependencies
- Ensuring consistent behavior across environments
- Making deployment faster and more reliable

Every DevOps engineer is expected to understand **Docker + Cloud VM deployment.**

What You Will Build

You will build and deploy a **Dockerized version of your portfolio website** that:

- Runs inside a Docker container
- Uses an Nginx web server
- Is deployed on a cloud virtual machine (AWS EC2 or equivalent)
- Is accessible using a public IP address

This setup closely mirrors **entry-level DevOps production environments**.

Concepts You Will Learn

Concept	Explanation
Docker Images	Blueprint of an application
Docker Containers	Running instance of an image
Dockerfile	Instructions to build an image
Port Mapping	Exposing container services
Cloud VM	Virtual server on the cloud
Manual Deployment	Foundation before CI/CD

Tools & Technologies Used

- Docker
- Nginx
- AWS EC2 (or any Linux-based cloud VM)
- Git & GitHub
- Linux terminal (basic commands)

Prerequisites

Before starting this task, ensure:

- You have completed Task 1 (static website ready)
- Docker is installed on your local system
- You have a basic understanding of Linux commands
- You have access to a cloud account (AWS / Azure / GCP)

Step-by-Step Implementation

Step 1: Prepare Your Website Files

Ensure your project folder contains:

```pgsql
project-folder/
│
├── index.html
├── styles.css
├── assets/
└── images/
```

This is the same website you created in Task 1.

Step 2: Install Docker

On Linux

```bash
sudo apt update
sudo apt install docker.io -y
sudo systemctl start docker
sudo systemctl enable docker
```

Verify installation:

```bash
docker --version
```

Step 3: Create a Dockerfile

Inside your project folder, create a file named **Dockerfile** (no extension).

```dockerfile
FROM nginx:alpine

COPY . /usr/share/nginx/html

EXPOSE 80
```

Explanation:

- `FROM nginx:alpine` → Uses a lightweight web server image
- `COPY` → Copies your website files into the container
- `EXPOSE 80` → Opens port 80 for HTTP traffic

Step 4: Build Docker Image

Run the following command inside your project folder:

```bash
docker build -t portfolio-website .
```

Verify image creation:

```bash
docker images
```

Step 5: Run the Docker Container Locally

```bash
docker run -d -p 8080:80 portfolio-website
```

Open browser:

```arduino
http://localhost:8080
```

At this stage, your website should be running **inside a Docker container**.

Cloud Deployment Phase

Step 6: Create a Cloud Virtual Machine

Using AWS EC2 as reference:

- Launch an Ubuntu instance
- Instance type: t2.micro (Free Tier)
- Allow inbound traffic on port 80
- Create or select a key pair

Step 7: Connect to the VM

```bash
ssh -i your-key.pem ubuntu@<PUBLIC_IP>
```

Step 8: Install Docker on the VM

```bash
sudo apt update
sudo apt install docker.io -y
sudo systemctl start docker
```

Optional:

```bash
sudo usermod -aG docker ubuntu
```

(Log out and log in again)

Step 9: Transfer Project Files to VM

Option 1: Clone from GitHub

```bash
git clone https://github.com/yourusername/portfolio-website.git
cd portfolio-website
```

Option 2: SCP from local system

```bash
scp -i your-key.pem -r project-folder ubuntu@<PUBLIC_IP>:/home/ubuntu/
```

Step 10: Build and Run Container on VM

```bash
docker build -t portfolio-website .
docker run -d -p 80:80 portfolio-website
```

Step 11: Access the Website

Open browser:

```cpp
http://<PUBLIC_IP>
```

Your Dockerized website is now live on the cloud.

Key Learning Outcomes

After completing Task 2, you will be able to:

- Explain what containers are and why they are used
- Write a basic Dockerfile
- Build and run Docker images
- Deploy containerized applications on cloud virtual machines
- Understand manual deployment workflows used before automation

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Deliverables

Mandatory:

1. Dockerfile
2. GitHub repository with source code
3. Screenshot of running container (docker ps)
4. Public IP URL showing live website
5. README explaining deployment steps

Optional:

- Short screen recording of deployment process
- Notes on issues faced and solutions

1) Docker — Official Documentation (Free)

- Docker Getting Started Guide
  https://docs.docker.com/get-started/
- Dockerfile Reference
  https://docs.docker.com/engine/reference/builder/
- Docker CLI Guide
  https://docs.docker.com/engine/reference/commandline/docker/

These are the authoritative resources that explain everything from basics to advanced usage.

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2) Free YouTube Tutorials (Docker)

Beginner-Friendly, Project-Oriented Playlists

Docker Full Course — FreeCodeCamp (4+ hours)

This covers Docker from scratch and includes building images and containers.
https://www.youtube.com/watch?v=fqMOX6JJhGo

Docker for Beginners (TechWorld with Nana)

Excellent visual, step-by-step approach for understanding concepts.
https://www.youtube.com/watch?v=3c-7n2ytqKs

Docker Tutorial for Beginners (Traversy Media)

Practical beginner Docker guide using real examples.
https://www.youtube.com/watch?v=pTFZFxd4hOI

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3) Cloud Virtual Machine (AWS EC2) — Official Guides

AWS EC2 Launch Instance (Step-by-Step)

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EC2_GetStarted.html

Authorize Security Groups for HTTP

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/authorizing-security-group-rules.html

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4) Free YouTube Tutorials (AWS EC2 Basics)

AWS EC2 Full Course (FreeCodeCamp)

Great beginner AWS EC2 guide, including SSH and instance setup.
https://www.youtube.com/watch?v=vjJwV8jfEoU

AWS EC2 Tutorial for Beginners (Simplilearn)

Short, direct, and easy to follow.
https://www.youtube.com/watch?v=3hFMYC1gKto

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5) Linux Basics (Required for VM Deployment)

Even if you are new to Linux, you will need basic knowledge of commands like scp, chmod, ssh, etc.

Linux for Beginners (FreeCodeCamp)

https://www.youtube.com/watch?v=IwPg-F1L7Fc

Linux Command Line Tutorial (Traversy Media)
https://www.youtube.com/watch?v=ZtqBQ68cfJc

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6) Git & GitHub — Official Free Learning
Git Handbook
https://guides.github.com/introduction/git-handbook/

GitHub Learning Lab (Interactive)
https://lab.github.com/

YouTube — Git & GitHub Crash Course (Traversy Media)
https://www.youtube.com/watch?v=SWYqp7iY_Tc

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7) Free Practice Platforms
These are excellent for hands-on command practice:
Play with Docker Playground (official sandbox)
https://labs.play-with-docker.com/

AWS Free Tier (12 months)
Provides free access to EC2, S3, Lambda, etc.
https://aws.amazon.com/free/
Katacoda Docker Scenarios
Interactive Docker practice
https://www.katacoda.com/courses/docker

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8) Cheat Sheets (Free)
These are extremely helpful as quick references while you code:
Docker Commands Cheat Sheet
https://dockerlabs.collabnix.com/docker/cheat-sheet/

AWS EC2 Cheat Sheet
https://tutorialsdojo.com/aws-cheat-sheet/

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9) Recommended Free Books / eBooks
Docker in Practice (Manning Free Chapter Excerpts)
https://www.manning.com/books/docker-in-practice
AWS Well-Architected Framework (Official)
https://aws.amazon.com/architecture/well-architected/

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10) Example Walkthroughs (Free Guides)
DigitalOcean Docker Tutorial (step-by-step)
https://www.digitalocean.com/community/tutorials/how-to-install-and-use-docker

Deploy Docker Containers on Ubuntu Server
https://www.digitalocean.com/community/tutorials/deploy-docker-containers-on-ubuntu

Recommended Learning Path (Order You Should Follow)

Step	Resource
1	Docker Official Getting Started
2	YouTube Docker Full Course (FreeCodeCamp)
3	Practice Play-with-Docker
4	Linux Basics YouTube Course
5	AWS EC2 Beginner Tutorial
6	Launch an EC2 Instance & SSH
7	Build Docker Image and Test Locally
8	Deploy Docker Container on EC2
9	Document Deployment Steps
10	Prepare Screenshots and Build README

Tools You Will Use (All Free)

Tool	Purpose
Docker Desktop	Build and run Docker locally
AWS EC2 Free Tier	Cloud VM deployment
VS Code	Code and Dockerfile editing
Terminal / PowerShell	Command execution
Git & GitHub	Version control and project storage
Browser	Access public endpoint